Jupyter Lab plug-in installation

Jupyter Lab plug-in installation
1. Install Node.js
2. Enable plug-in
3. Install the expansion pack

Installing plugins can enhance the functionality of Jupyter Lab and provide more tools and features.

1. Install Node.js

- View the architecture of Raspberry Pi

```
uname -a
```

```
yahboom@raspberrypi:~ $ uname -a
Linux raspberrypi 6.1.0-rpi7-rpi-2712 #1 SMP PREEMPT Debian 1:6.1.63-1+rpt1 (2023-11-24) aarch64 GNU/Linux
```

Select the Node.js version suitable for ARMv8 architecture to install

- Node.js download

```
https://nodejs.org/en/download/
```

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Downloads

Latest LTS Version: **20.10.0** (includes npm 10.2.3)

Download the Node.js source code or a pre-built installer for your platform, and start developing today.

LTS Recommended For Most Users	Current Latest Features

Windows Installer	macOS Installer	Source Code
node-v20.10.0-x64.msi	node-v20.10.0.pkg	node-v20.10.0.tar.gz

Windows Installer (.msi)	32-bit	64-bit	ARM64
Windows Binary (.zip)	32-bit	64-bit	ARM64
macOS Installer (.pkg)	64-bit / ARM64		
macOS Binary (.tar.gz)	64-bit	ARM64	
Linux Binaries (x64)	64-bit		
Linux Binaries (ARM)	ARMv7	ARMv8	
Source Code	node-v20.10.0.tar.gz		

Additional Platforms

Docker Image	Official Node.js Docker Image
Linux on Power LE Systems	64-bit
Linux on System z	64-bit
AIX on Power Systems	64-bit

- Unzip

Download the latest version of the binary file from the official website, unzip it, and name the folder the node folder:

```
cd Downloads/
tar -xJf node-v20.10.0-linux-arm64.tar.xz
mv -i node-v20.10.0-linux-arm64 node
```

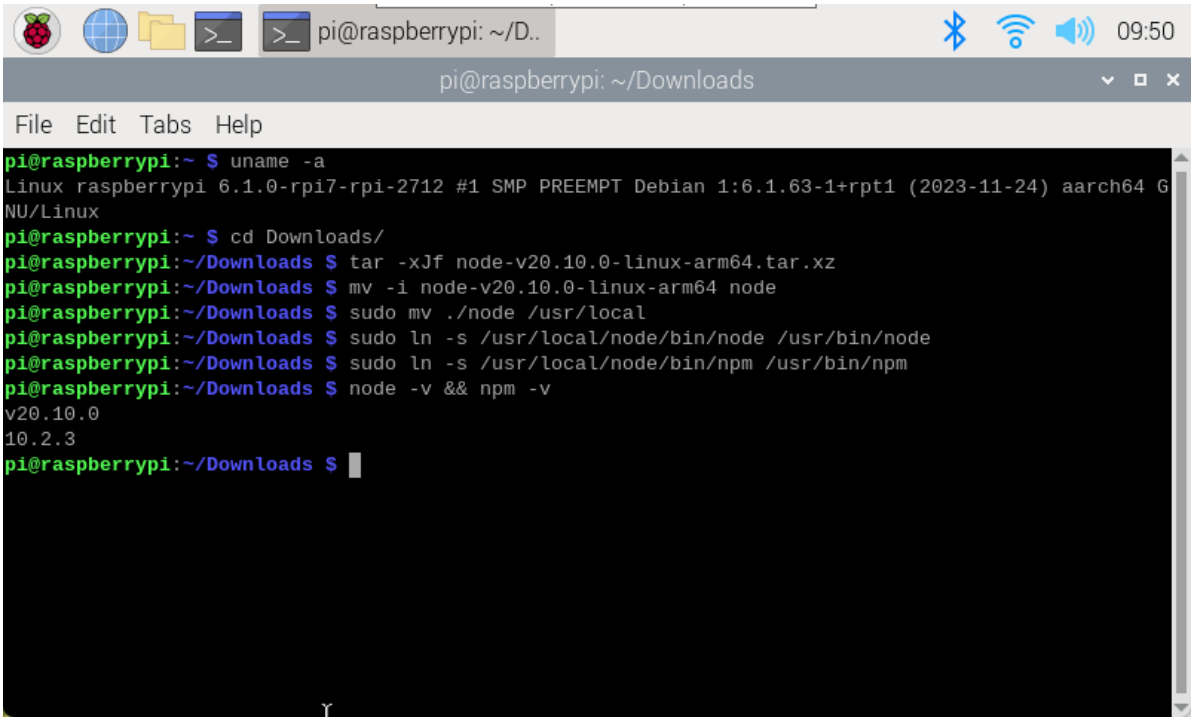
```
yahboom@raspberrypi:~ $ cd Downloads/
yahboom@raspberrypi:~/Downloads $ ls
node-v20.10.0-linux-arm64.tar.xz
yahboom@raspberrypi:~/Downloads $ tar -xJf node-v20.10.0-linux-arm64.tar.xz
yahboom@raspberrypi:~/Downloads $ mv -i node-v20.10.0-linux-arm64 node
yahboom@raspberrypi:~/Downloads $ ls
node  node-v20.10.0-linux-arm64.tar.xz
```

- Move binaries and add soft links

```
sudo mv ./node /usr/local
sudo ln -s /usr/local/node/bin/node /usr/bin/node
sudo ln -s /usr/local/node/bin/npm /usr/bin/npm
```

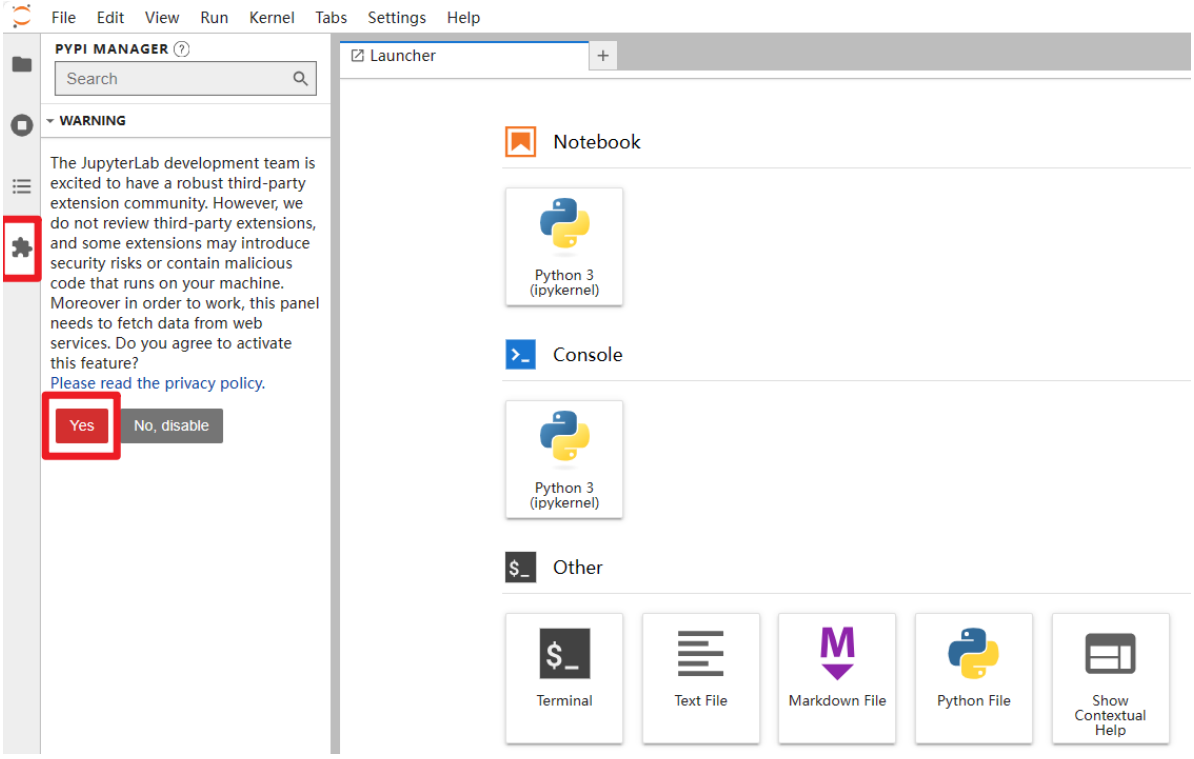
- Check whether node and npm are installed successfully

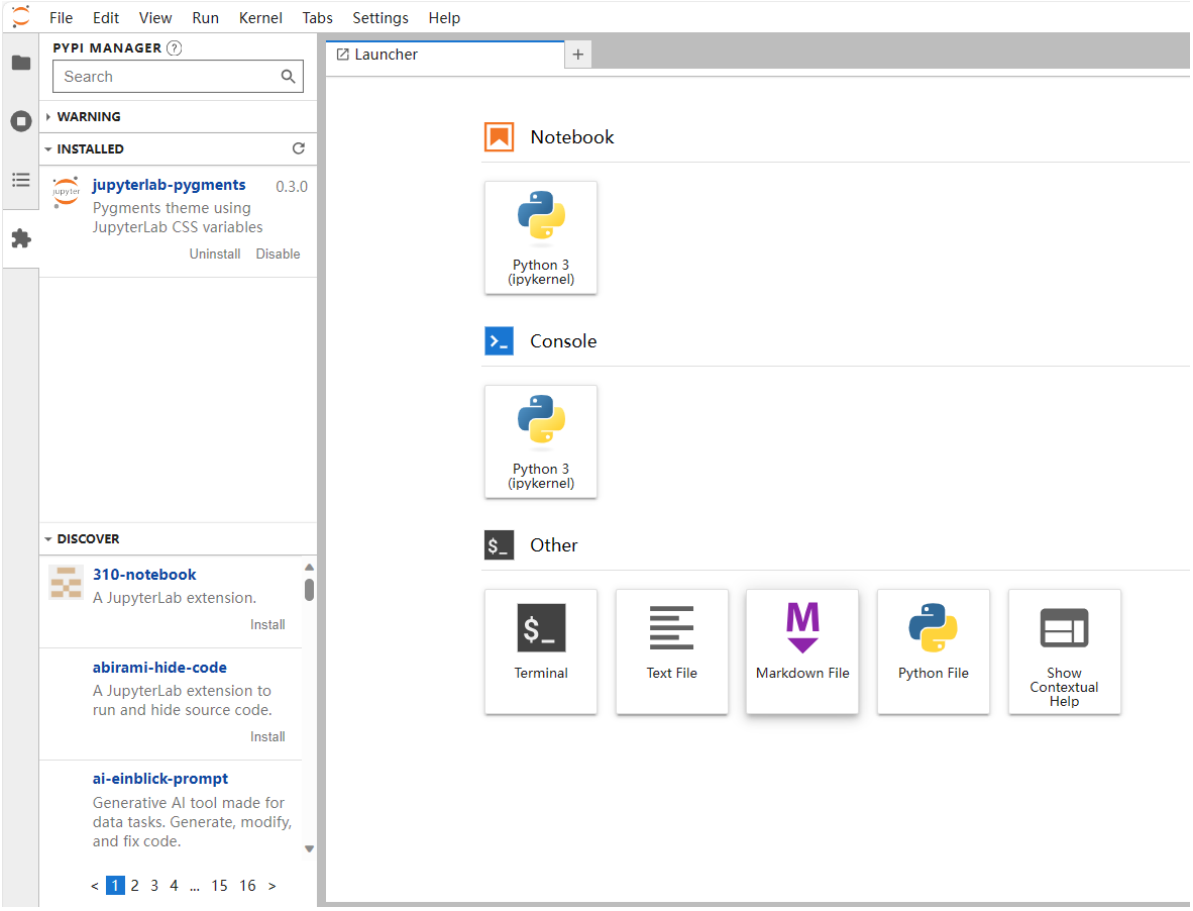
```
node -v && npm -v
```



2. Enable plug-in

Enable third-party expansion support in jupyter lab.

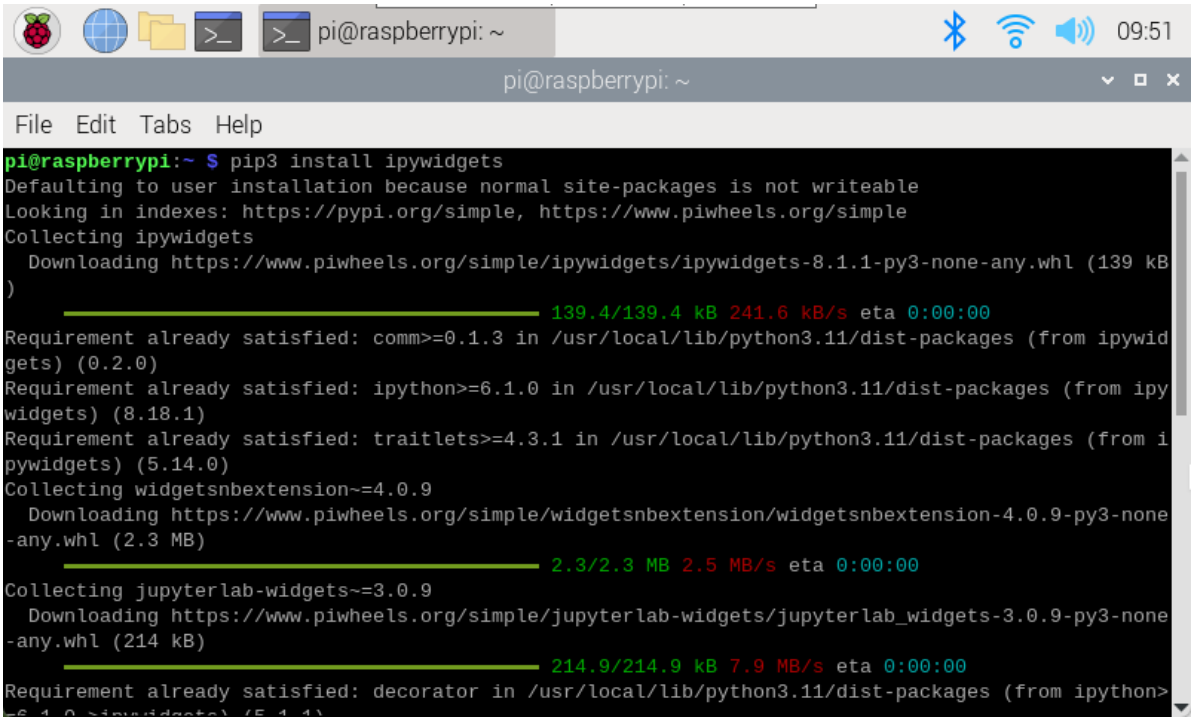


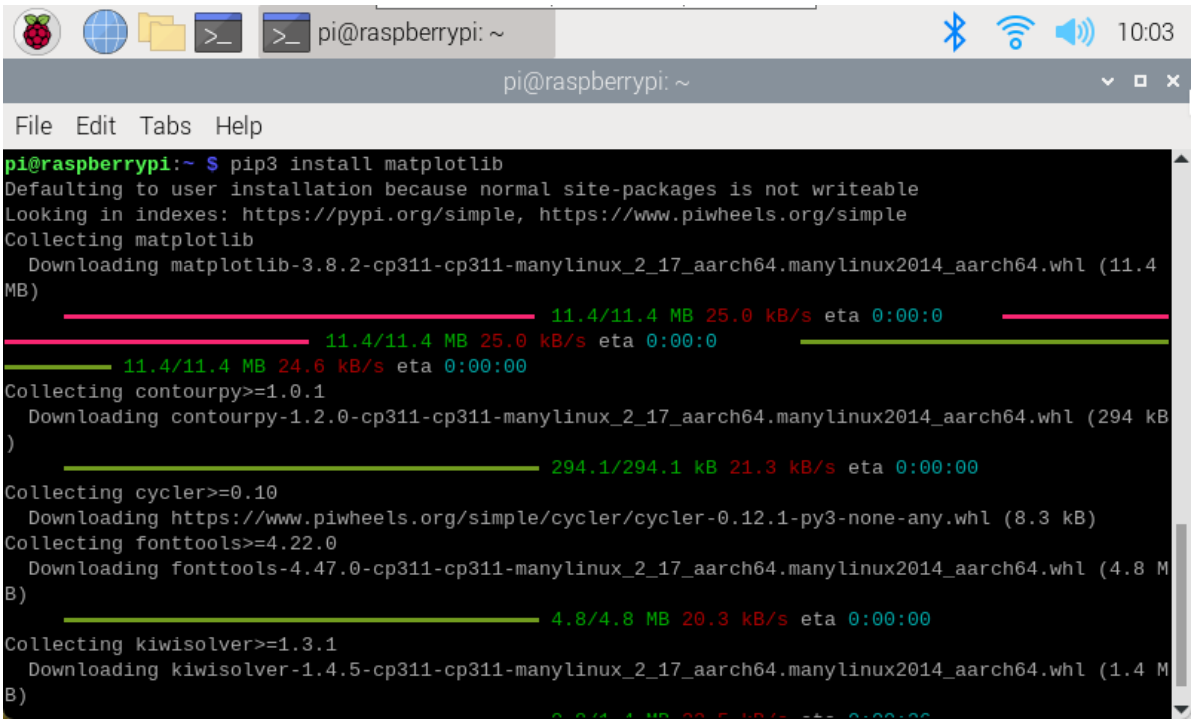


3. Install the expansion pack

```
pip3 install ipywidgets
```

Provides functionality for creating interactive widgets in Jupyter Lab.




```
pip3 install matplotlib
```

Features for creating various types of graphs and visualizations in Python, including line plots, scatter plots, histograms, and more.